

$$y = mx + b$$

Use points (1, 3) and (3, 6)

$$m = -1 \quad \alpha = 0.1$$

$$b = 1$$

predicted value $\bar{y} = mx + b$

Mean Squared Error gradient $\frac{\Delta E}{\Delta m} = -\frac{2}{n} \sum (y - y_i) x_i$

$$\frac{\Delta E}{\Delta b} = -\frac{2}{n} \sum (y - y_i)$$

We know that $M_{\text{new}} = M_{\text{old}} - \alpha \frac{\Delta E}{\Delta m}$

$$B_{\text{new}} = B_{\text{old}} - \alpha \frac{\Delta E}{\Delta b}$$

First Iteration

$$m = -1 \quad b = 1 \quad \alpha = 0.1 \quad (3, 6)$$

for (1, 3) $y_1 = (-1)(1) + 1 = 0$

$y_2 = (-1)(3) + 1 = -2$

(0, -2) → predicted values

Error $E = y - \bar{y} \quad \begin{cases} 3 - 0 = 3 \\ 6 - (-2) = 8 \end{cases} \quad (3, 8)$

$$\frac{\Delta E}{\Delta m} = -\frac{2}{n} \sum (y - y_i) x_i (=) -\frac{2}{2} [1(3-0) + 3(6+2)] = -27$$

$$\frac{\Delta E}{\Delta b} = -\frac{2}{n} \sum (y - y_i) (=) -\frac{2}{2} [3+8] = -11$$

So the update $M_{\text{new}} = M_{\text{old}} - \alpha \frac{\Delta E}{\Delta m} (=) (-1) - 0.1(-27)$

$$m_{\text{new}} (=) \underline{\underline{1.7}}$$

~~new~~
 $b_{\text{new}} = b_{\text{old}} - \alpha \frac{\Delta E}{\Delta b} (=) 1 - 0.1(11) = \underline{\underline{2.1}}$